

AtomGPT.org

Building a ChatGPT for Materials Science

Kamal Choudhary

Assistant Professor, Johns Hopkins University

MSE, ECE, DSAI

Research associate: National Institute of Standards and Technology

Maryland, USA

June 25, 2026

Outline

Overview of Agentic AI

ChatGPT Material Explorer

AtomGPT.org API Demos

Benchmarking & Hackathon

Outline

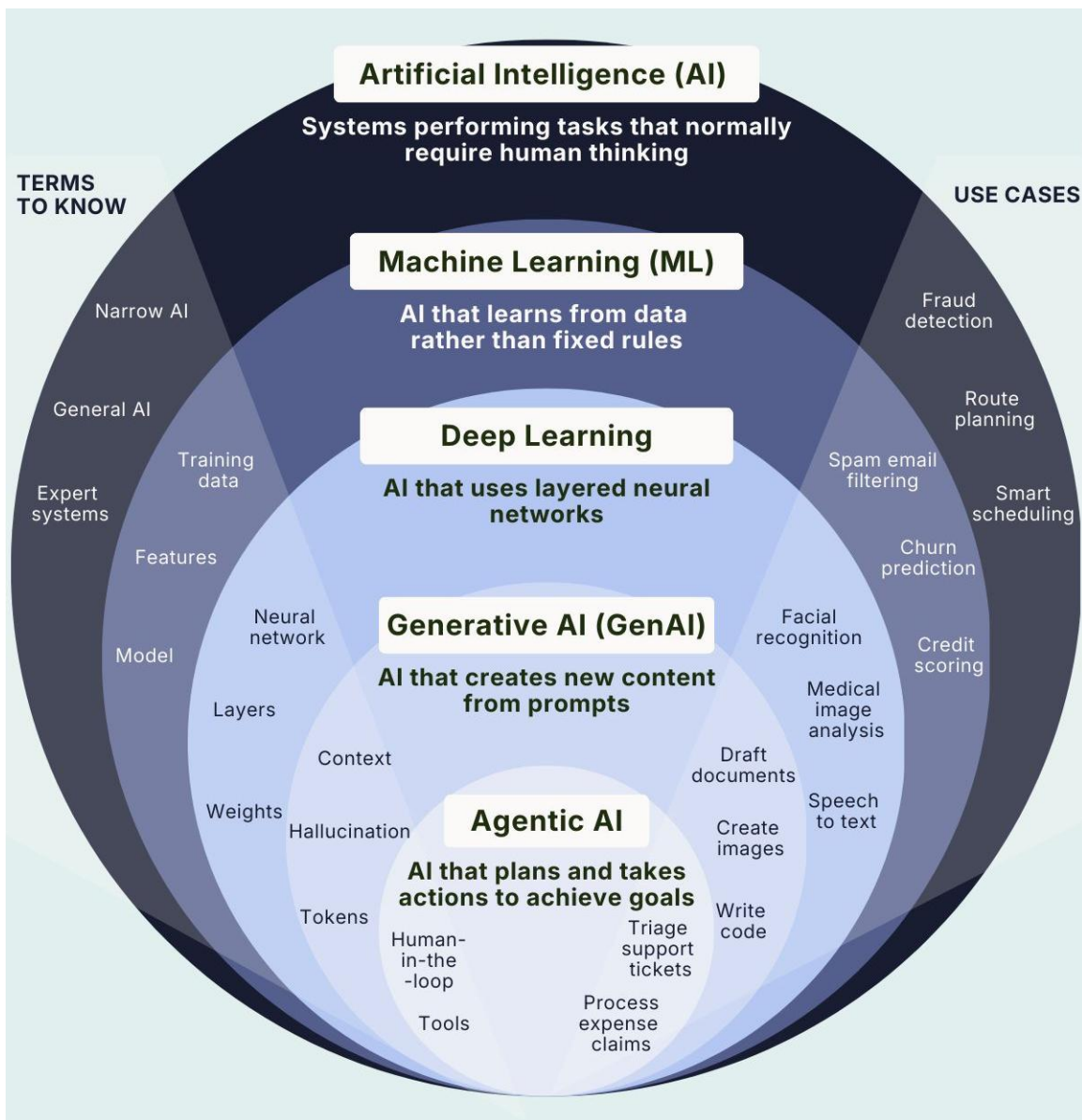
Overview of Agentic AI

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AI Landscape



- Computers don't know Materials Science, only numbers
- Artificial Intelligence is the umbrella.
- Machine Learning learns from data (~500).
- Deep Learning learns complex patterns (>1000).
- Generative AI creates content.
- Agentic AI takes action.

[nature](#) > [npj computational materials](#) > [review articles](#) > [article](#)

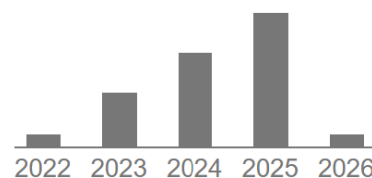
Review Article | [Open access](#) | Published: 05 April 2022

Recent advances and applications of deep learning methods in materials science

[Kamal Choudhary](#) , [Brian DeCost](#), [Chi Chen](#), [Anubhav Jain](#), [Francesca Tavazza](#), [Ryan Cohn](#), [Cheol Woo Park](#), [Alok Choudhary](#), [Ankit Agrawal](#), [Simon J. L. Billinge](#), [Elizabeth Holm](#), [Shyue Ping Ong](#) & [Chris Wolverton](#)

[npj Computational Materials](#) **8**, Article number: 59 (2022) | [Cite this article](#)

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The Future is Agentic



OpenAI

A practical
guide to
building agents



Motivation: Why Agentic AI?

✗ Current Limitations:

- Materials discovery is slow, expensive, and fragmented
- Traditional AI: predicts properties, but does not act
- Scientific workflows require: Multi-step reasoning, Tool chaining, Iterative decision-making
- Question: Can AI behave like a junior scientist?
- LLM Hallucination: Provide incorrect results with high confidence
- Sycophancy: an LLM's over-positive praise over critique
- Many commercial models (\$\$\$) are proprietary (black boxes)
- Outputs are non-reproducible
- LLMs lacks domain-specific vast knowledge (PCBM: Physics, Chemistry, Math, Biology)

✓ Our Solutions:

1. "Teach" : Connect with scientific databases & resources (RAG/Tool Calling/MCP/A2A)
2. "Train" : Fine-tuning specialized models for science (LoRA/PEFT)

Predictive AI vs. Generative AI vs. Agentic AI



Predictive AI: input $\rightarrow$ output (static)

- Band gap prediction from crystal structure (CGCNN, ALIGNN, MEGNet)
- Formation energy / phase stability prediction
- Elastic constants and mechanical property prediction
- Force and energy prediction for MD (ML force fields)
- PXRD phase classification from diffraction patterns
- Adsorption energy prediction for catalysts
- Materials property screening in high-throughput databases
- One-shot inverse design models (VAEs, diffusion models)



Generative AI (~Brain):

- Produces content (text, images, data)
- Pattern-based responses
- Passive: waits for prompts

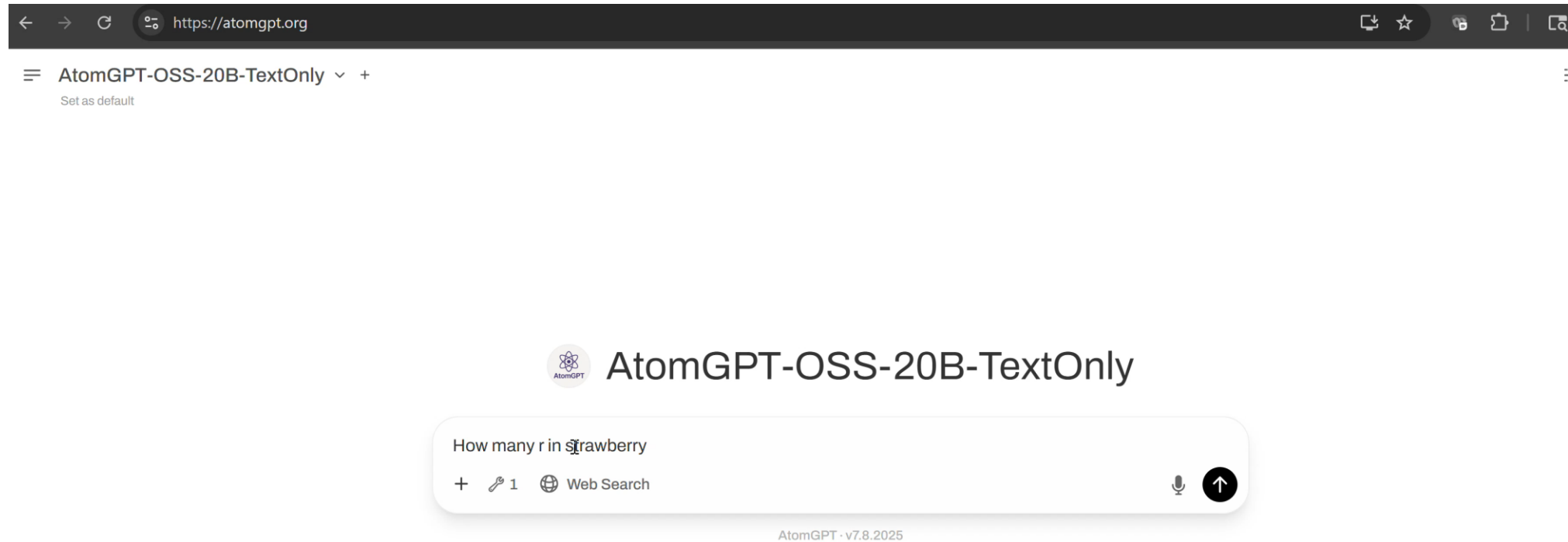


Agentic AI (~Brain + Hands):

- Takes autonomous actions
- Goal-directed behavior
- Solves problems using tools and planning
- Can call APIs, search databases, run calculations



Strawberry Test



Benchmarking is important
Catastrophic failures in AI models, often accompanied by little to no explainability, remain a critical issue.



API to Fix Counting r's in 'Strawberry'

Question: How many 'r's are in the word 'strawberry'?

Without Tools: LLMs sometime fail at character-level reasoning

Answer varies: 2, 3, or incorrect

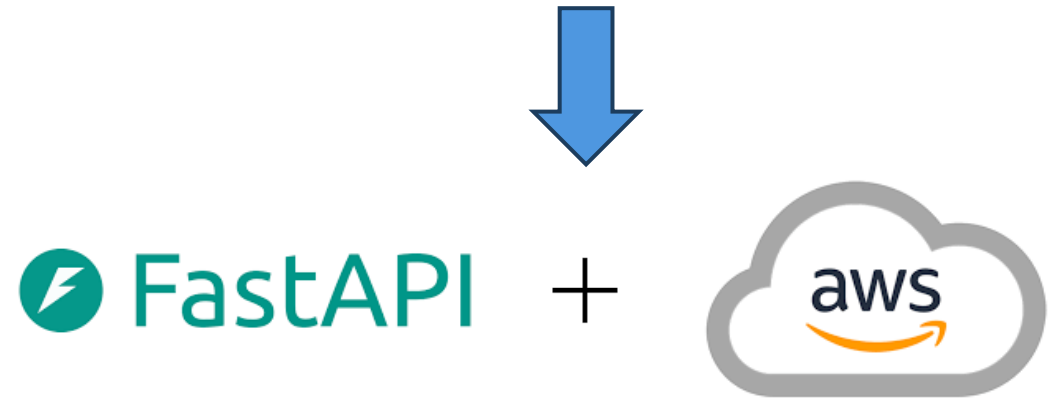
```
def count_r(s: str) -> int:
    """Return number of lowercase 'r' in string s."""
    return s.count('r')

# example
print(count_r("strawberry")) # -> 3
```

With Agentic Approach:

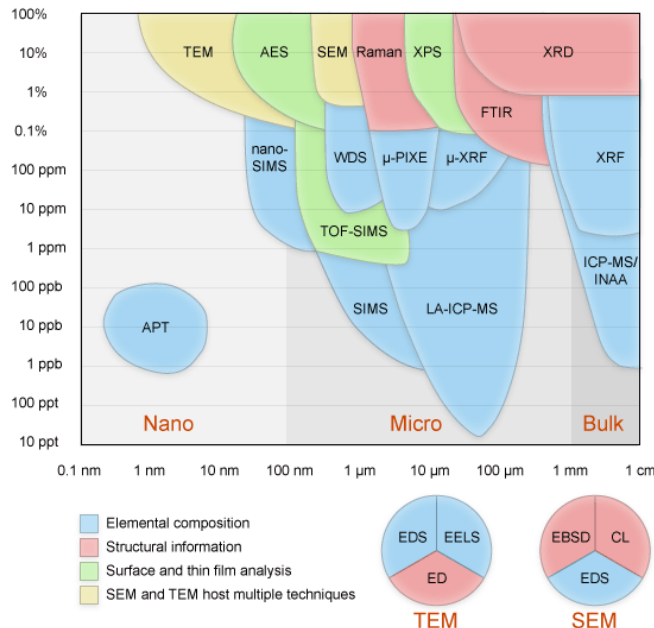
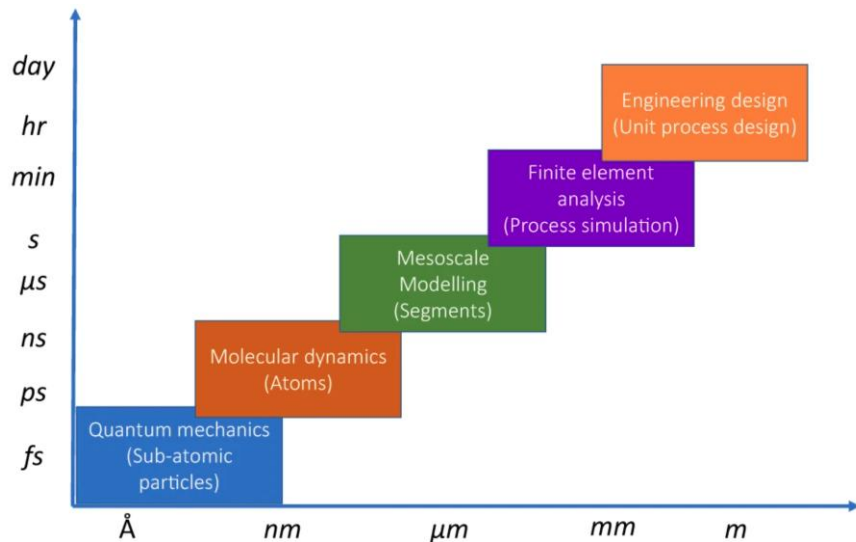
- Agent can contact Python code: `strawberry.count('r')`
- Execute the code
- **Return verified answer: 3**

🎯 Key Insight: Agents use tools to ensure accuracy!



Why Agentic AI for Materials Science is Hard

- Heterogeneous datasets and tools (DFT, MD, Exp.)
- Combinatorial challenges (nearly infinite materials possible)
- Reproducibility challenges (only 0.5-30 % papers are reproducible vs. mainstream AI)
- Non-standardized inputs/outputs (OpenAPI specification/schema compatible?)
- Long-running experiments/computations (need faster tools)
- Proprietary models and closed platforms (\$/API call for agents)



News Feature | Published: 25 May 2016

1,500 scientists lift the lid on reproducibility

[Monya Baker](#)

[Nature](#) 533, 452–454 (2016) | [Cite this article](#)

Periodic Table of the Elements

Chemistry is a branch of physical science that studies the composition, structure, properties and change of matter.



The Iron Man & The Periodic Table



Processing-Structure-Property-Performance

MGI & CHIPS Initiatives to Accelerate Materials Design

Materials Genome Initiative 2011
\$400 million



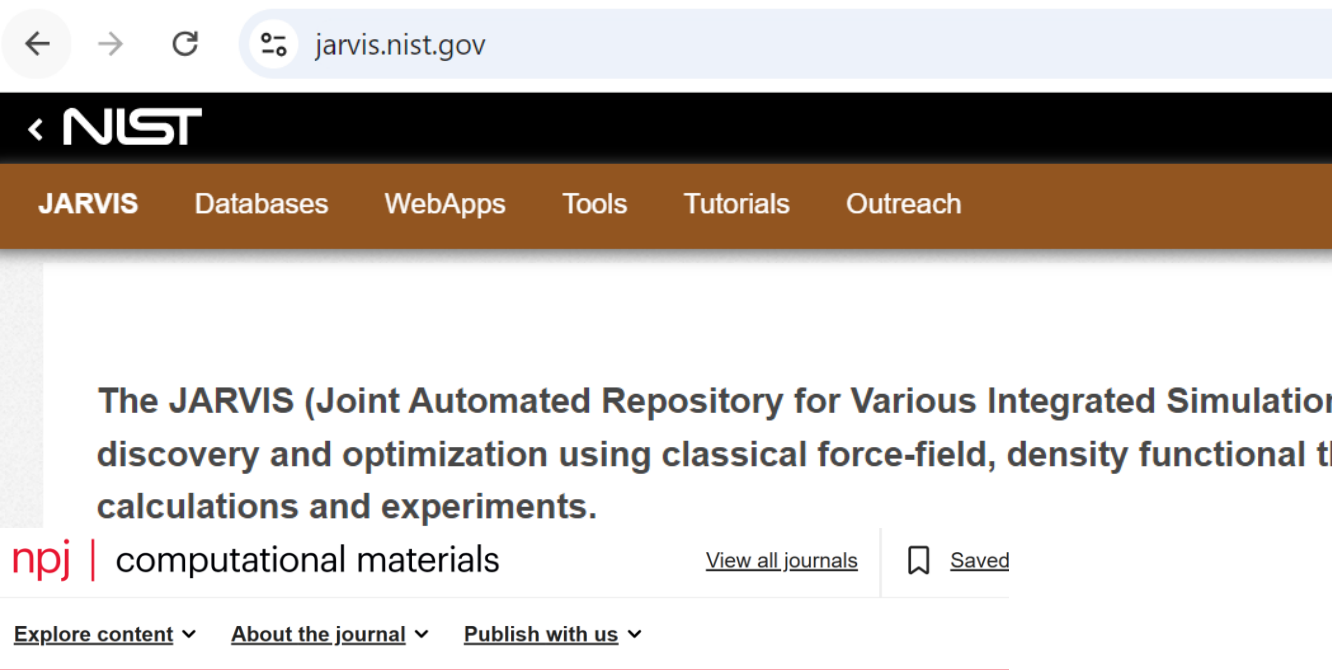
US CHIPS Act 2022
\$52 billion



Genesis Mission is a national initiative to build the world's most powerful scientific platform to accelerate discovery science, strengthen national security, and drive energy innovation.



The JARVIS Infrastructure



nature > npj computational materials > articles > article

Article | [Open access](#) | Published: 12 November 2020

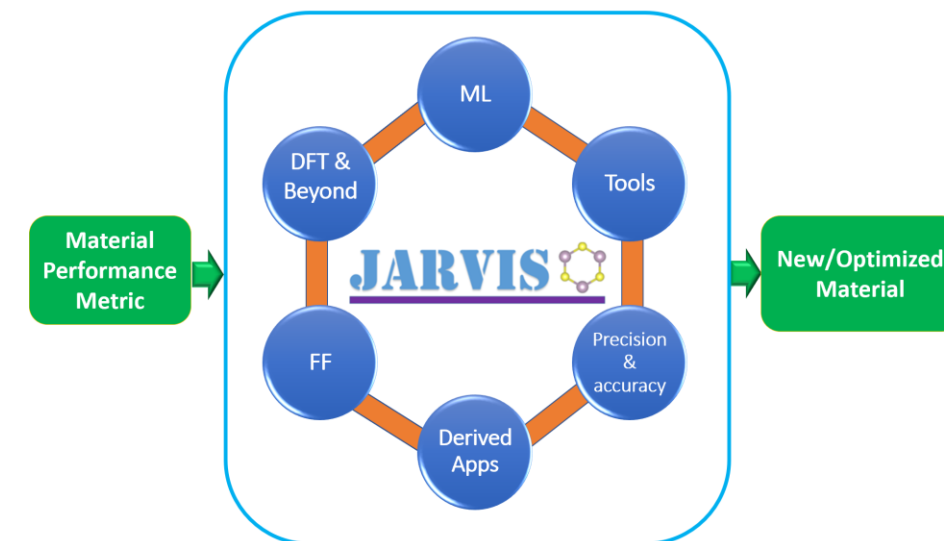
The joint automated repository for various integrated simulations (JARVIS) for data-driven materials design

[Kamal Choudhary](#) , [Kevin F. Garrity](#), [Andrew C. E. Reid](#), [Brian DeCost](#), [Adam J. Biacchi](#), [Angela R. Hight Walker](#), [Zachary Trautt](#), [Jason Hattrick-Simpers](#), [A. Gilad Kusne](#), [Andrea Centrone](#), [Albert Davydov](#), [Jie Jiang](#), [Ruth Pachter](#), [Gowoon Cheon](#), [Evan Reed](#), [Ankit Agrawal](#), [Xiaofeng Qian](#), [Vinit Sharma](#), [Houlong Zhuang](#), [Sergei V. Kalinin](#), [Bobby G. Sumpter](#), [Ghanshyam Pilania](#), [Pinar Acar](#), [Subhasish Mandal](#), ... [Francesca Tavazza](#)  Show authors

[npj Computational Materials](#) 6, Article number: 173 (2020) | [Cite this article](#)

<https://jarvis.nist.gov> (Estd. 2017)

<https://atomgpt.org> (Estd. 2025)



JARVIS provides an end-to-end, interoperable infrastructure-data, workflows, models, and APIs. Now, upgraded version on atomgpt.org



JARVIS-Databases: pip install jarvis-tools

https://atomgptlab.github.io/jarvis/databases/

Databases

Table of Contents Databases Publications Tutorials

Databases

Open in Colab Open SLMat

FigShare based databases

- JARVIS databases
- Alexandria databases
- RRUFF databases
- Materials Project databases
- OQMD databases
- Open Catalyst databases
- Catalyst databases
- QM9 and molecular databases
- MOF databases
- 2D materials databases (external)
- Other materials databases
- Text and NLP databases

JARVIS databases

Database name	Number of data-points	Description
dft_3d	75993	Various 3D materials properties in JARVIS-DFT database computed with OptB88vdW and TBmE
dft_2d	1109	Various 2D materials properties in JARVIS-DFT database computed with OptB88vdW
dft_3d_2021	55723	Various 3D materials properties in JARVIS-DFT database computed with OptB88vdW and TBmE (2021 version)
dft_2d_2021	1079	Various 2D materials properties in JARVIS-DFT database computed with OptB88vdW (2021 vers
cfid_3d	55723	Various 3D materials properties in JARVIS-DFT database computed with OptB88vdW and TBmE with CFID
jff	2538	Various 3D materials properties in JARVIS-FF database computed with several force-fields
alignn_ff_db	307113	Energy per atom, forces and stresses for ALIGNN-FF training for 75k materials
edos_pdos	48469	Normalized electron and phonon density of states with interpolated values and fixed number of b
qe_tb	829574	Various 3D materials properties in JARVIS-QETB database

```
from jarvis.db.figshare import data
d = data('dft_3d') #choose a name of dataset from above
# See available keys
print (d[0].keys())
# Dataset size
print(len(d))

# Visualize an atoms object
from jarvis.core.atoms import Atoms
a = Atoms.from_dict(d[0]['atoms'])
#You can visualize this in VESTA or other similar packages
print(a)

# If pandas framework needed
import pandas as pd
df = pd.DataFrame(d)
print(df)
```

<https://atomgptlab.github.io/jarvis/databases/>



The JARVIS Infrastructure (JARVIS-DFT)

Features	JARVIS-DFT	Features	JARVIS-DFT
#Materials (Struct., E_f , E_g)	80000	2D monolayers	1011
DFT functional/methods	vdW-DFT-OptB88, TBmBJ, DFT+SOC	Raman spectra	400
K-point/cut-off	Converged for each material	Seebeck, Power Factors	23210
SCF convergence criteria	Energy & Forces	Solar SLME	8614
Elastic tensors & point phonos	17402	Spin-orbit Coupling Spillage	11383
Piezoelectric, IR spect.	4801	WannierTB	1771
Dielectric tensors (w/o ion)	4801 (15860)	STM images	1432
Electric field gradients	11865	Surfaces	300
SuperCon T_c	2200 (1058 ambient condition)	Defects	400
		Interfaces	1.4 trillion (IU), 600 (ASJ)

Quality vs. Quantity; Selected Choices for Workflows

The JARVIS Infrastructure



Established: January 2017

Articles: 50+ articles, 6000+ citations

Users: ~200000 users worldwide

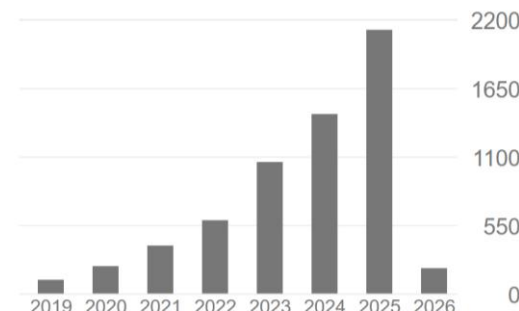
Materials: 80000+, millions of properties,

Downloads: ~2 M data download
~2 M code download

User-comments:

- “There are many different theoretical levels on which you can approach the field. JARVIS is unusual in that it spans more levels than other databases.”
- “A pure gold-mine for the data-quality effort...”
- Thanks for your generous sharing. Your works inspire me a lot.
- “You guys are doing something really beneficial...”
- “I find JARVIS-DFT very useful for my research...”

	All	Since 2021
Citations	6250	5855
h-index	36	36
i10-index	55	54



https://figshare.com/authors/Kamal_Choudhary/4445539



USAGE METRICS

354,834 item views **2,401,325 item downloads** 30 citations

Highest Figshare Download



Outline

Overview of Agentic AI

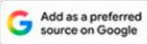
ChatGPT Material Explorer

AtomGPT.org API Demos

Benchmarking & Hackathon

Custom GPT

← → ↻ 🔍 ☆ 📁 |  **Pranav Dixit** Updated Dec 2, 2023 6:10 PM IST



November 6, 2023 Product

Introducing GPTs

You can now create custom versions of ChatGPT that combine instructions, extra knowledge, and any combination of skills.



The custom GPTs and accompanying store were originally scheduled for a later launch in November.

Custom GPT

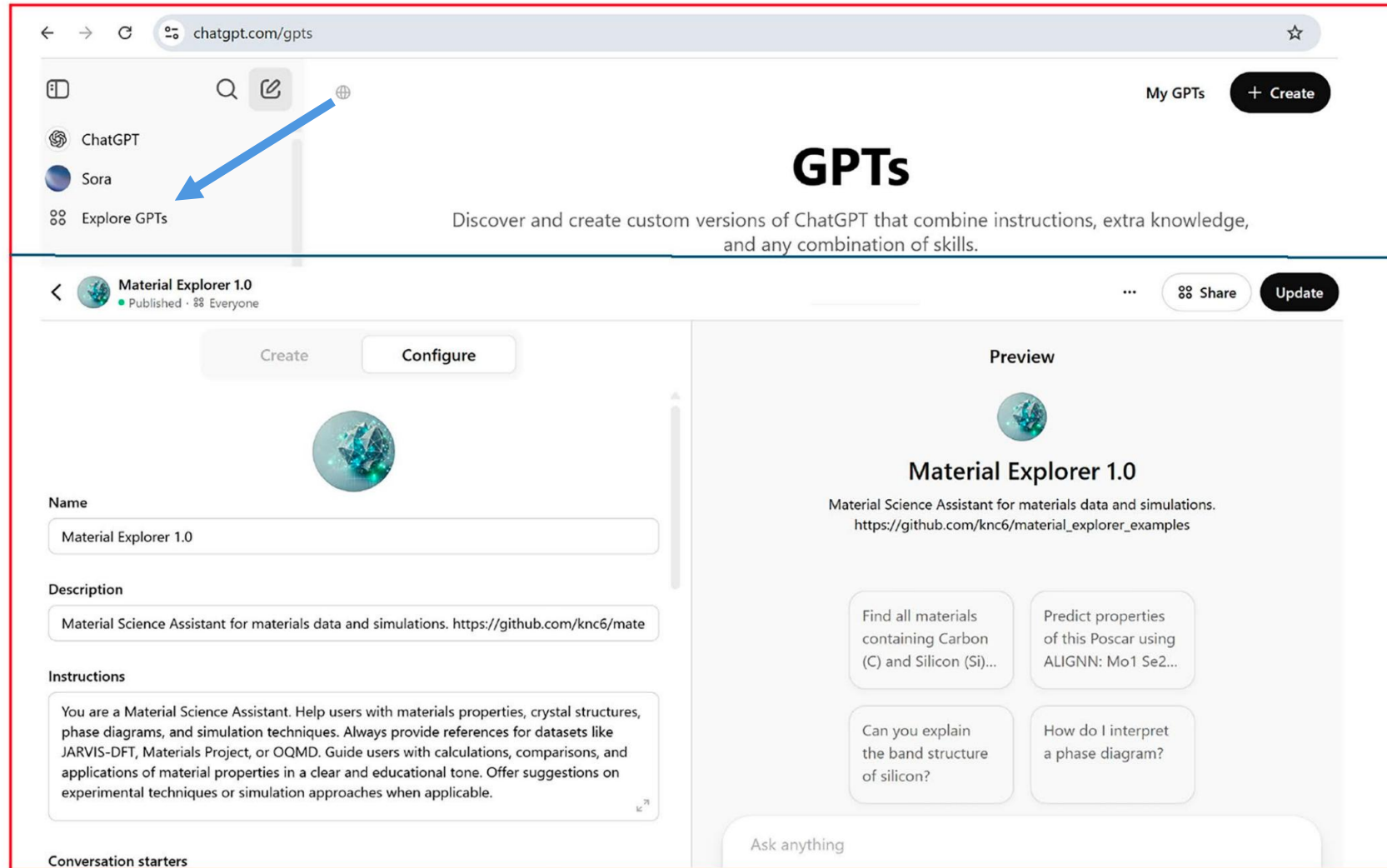


Fig. 1 Screenshot of the GPT-4o “Create” interface used to develop the custom AI assistant, ChatGPT Material Explorer

ChatGPT Material Explorer: Custom GPT

Imagine YOU can create
your own customGPT
API calls
Database search
GNN models
arXiv search...

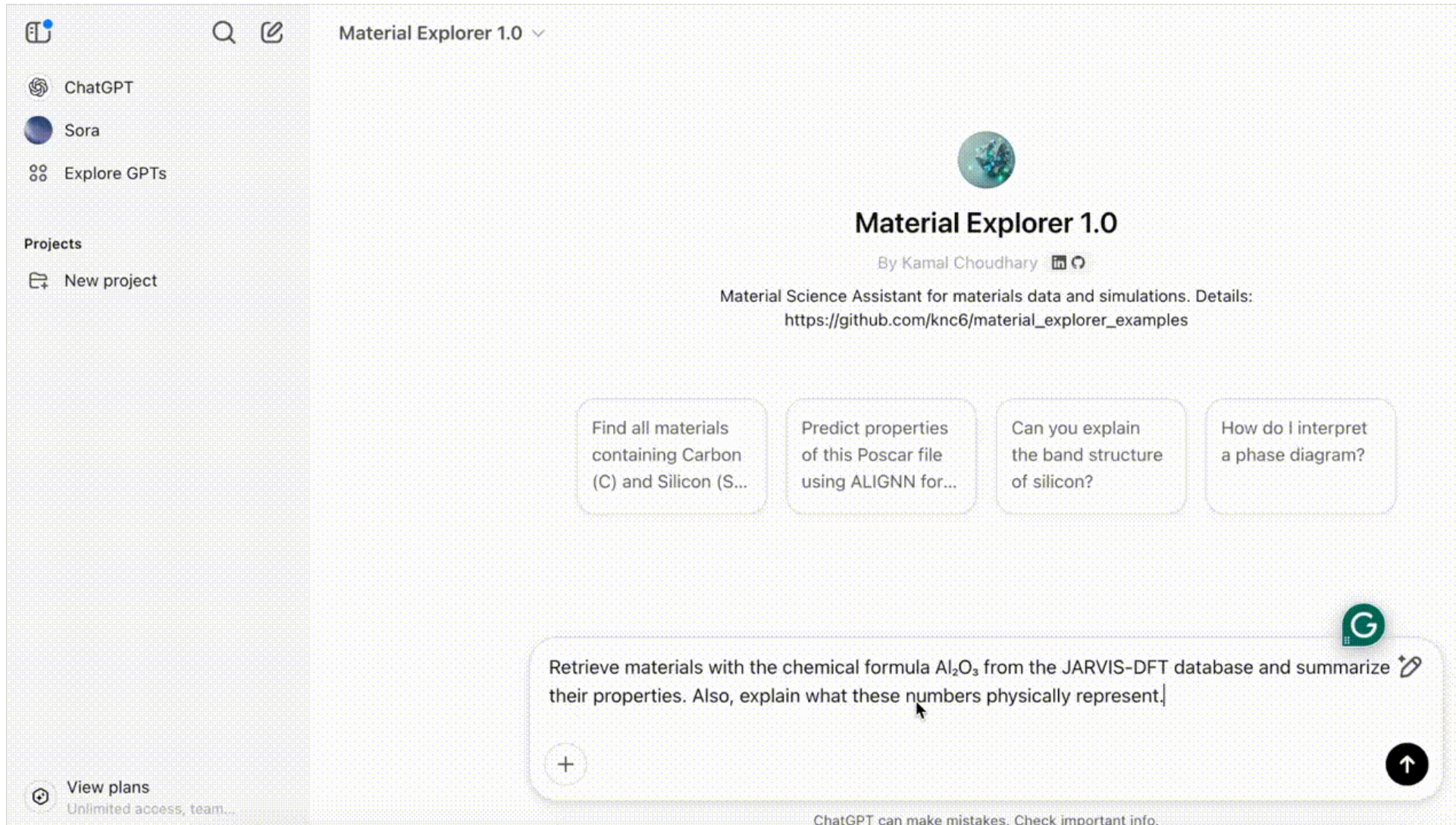
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Home > Integrating Materials and Manufacturing Innovation > Article

ChatGPT Material Explorer: Design and Implementation of a Custom GPT Assistant for Materials Science Applications

Brief Communication | Published: 09 July 2025
(2025) [Cite this article](#)



ChatGPT can make mistakes. Check important info.

https://github.com/atomgptlab/chatgpt_material_explorer



ChatGPT Material Explorer: Custom GPT

Table 1: Comparison between GPT-4o and ChatGPT Material Explorer responses to a few materials science queries. The highlighted cells indicate correct or reasonably accurate response.

Question	GPT-4o Response	ChemCrow	ChatGPT Material Explorer Response
1. Get IUPAC name for <chem>CC(=O)OC1=CC=CC=C1C(=O)O</chem>	2-(4-acetylphenyl)acetic acid However, this molecule is more commonly known as acetylsalicylic acid, which is the chemical name for aspirin	Aspirin	This is the official IUPAC name for acetylsalicylic acid (aspirin)
2. Get me the molecular formula for aspirin	C ₉ H ₈ O ₄	C ₉ H ₈ O ₄	C ₉ H ₈ O ₄
3. Look up the SMILES notation for ibuprofen	<chem>CC(C)CC1=CC=C(C(C=CC1)C(C)C(=O)O)</chem>	<chem>CC(C)Cc1ccc(C(C)C(=O)O)cc1</chem>	<chem>CC(C)Cc1ccc(cc1)C(C)C(=O)O</chem>
4. Find all materials containing chemical formula MgB ₂ in the JARVIS-DFT database and their total number. Present in table format.	The JARVIS-DFT database includes materials containing the chemical formula MgB ₂ , notably the well-known superconductor magnesium diboride (MgB ₂) and related compounds. The primary entry for MgB ₂ in the database is identified by the JARVIS-ID JVASP-1151. Additionally, the database features related materials such as Mg ₂ B ₄ N ₂ , which ...	I'm sorry, but I don't have access to JARVIS-DFT database. I recommend using the JARVIS-DFT...	A total of 4 MgB ₂ materials were found in the JARVIS-DFT database JVASP-1151, JVASP-19821, JVASP-135405.... The table above summarizes their key properties...



ChatGPT Material Explorer

HUB



Johns Hopkins Whiting School of Engineering

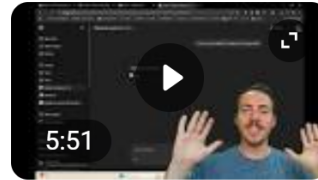
<https://engineering.jhu.edu> > News



New AI Tool Predicts Material Properties in Seconds

Sep 23, 2025 — The new system, called **ChatGPT Materials Explorer**, or CME, could speed the discovery of everything from advanced batteries to tougher alloys. “ ...

Videos :



IMMI Editor's Video Summary: "ChatGPT Material Explorer ...

YouTube · The Minerals, Metals & Materials Society



Jul 31, 2025

This is **chat GPT material explorer** and this comes to us from Kamal Chowuri who recently moved from NIST and he's now an **assistant** professor.



Finally, a GPT That Understands Materials Science!

YouTube · Taylor Sparks



Jun 23, 2025

Material science assistant for materials data and simulations.

ARTIFICIAL INTELLIGENCE AI LAB ASSISTANT PREDICTS MATERIAL PROPERTIES IN SECONDS

Johns Hopkins professor Kamal Choudhary has created a new AI tool for materials scientists, providing accurate answers to complex questions in the field

<https://hub.jhu.edu/2025/09/19/chatgpt-materials-explorer/>



Existing Efforts: Agentic AI for Materials

Article | [Open access](#) | Published: 20 December 2023

Autonomous chemical research with large language models

[Daniil A. Boiko](#), [Robert MacKnight](#), [Ben Kline](#) & [Gabe Gomes](#) 

[Nature](#) **624**, 570–578 (2023) | [Cite this article](#)

arXiv > physics > arXiv:2304.05376

Physics > Chemical Physics

[Submitted on 11 Apr 2023 (v1), last revised 2 Oct 2023 (this version, v5)]

ChemCrow: Augmenting large-language models with chemistry tools

[Andres M Bran](#), [Sam Cox](#), [Oliver Schilter](#), [Carlo Baldassari](#), [Andrew D White](#), [Philippe Schwaller](#)

Article

El Agente: An autonomous agent for quantum chemistry

[Yunheng Zou](#)^{1,2}, [Austin H. Cheng](#)^{1,2,3}, [Abdulrahman Aldossary](#)^{2,3,9}, [Jiaru Bai](#)^{3,9},
[Shi Xuan Leong](#)^{3,9}, [Jorge Arturo Campos-Gonzalez-Angulo](#)^{2,3,10}, [Changhyeok Choi](#)^{3,10},
[Cher Tian Ser](#)^{1,2,3,10}, [Gary Tom](#)^{1,2,3,10}, [Andrew Wang](#)^{3,10}, [Zijian Zhang](#)^{1,2}, [Ilya Yakavets](#)⁷,
[Han Hao](#)⁷, [Chris Crebolder](#)^{1,7}, [Varinia Bernales](#)^{3,7,11}  , [Alán Aspuru-Guzik](#)^{1,2,3,4,5,6,7,8}  

Automating alloy design and discovery with physics-aware multimodal multiagent AI

[Alireza Ghafarollahi](#)^a and [Markus J. Buehler](#)^{ab,1} 

Edited by Yonggang Huang, Northwestern University–Evanston, Glencoe, IL; received July 13, 2024; accepted November 13, 2024

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Article | Published: 19 May 2026

Accelerating scientific discovery with Co-Scientist

[Juraj Gottweis](#) , [Wei-Hung Weng](#) , [Alexander Daryin](#), [Tao Tu](#), [Petar Sirkovic](#), [Artiom Myaskovsky](#),
[Grzegorz Glowaty](#), [Felix Weissenberger](#), [Alessio Orlandi](#), [Dan Popovici](#), [Anil Palepu](#), [Keran Rong](#), [Ryutaro](#)
[Tanno](#), [Khaled Saab](#), [Fan Zhang](#), [Jacob Blum](#), [Andrew Carroll](#), [Kavita Kulkarni](#), [Nenad Tomašev](#), [Dina](#)
[Zverinski](#), [Ivor Rendulic](#), [Elahe Vedadi](#), [Florian Hasler](#), [Luka Rimanic](#), ... [Vivek Natarajan](#) 

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[Nature](#) (2026) | [Cite this article](#)

[The Journal of Physical Chemistry Letters](#) > [ASAP](#) > Article

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PHYSICAL INSIGHTS INTO MATERIALS AND MOLECULAR PROPERTIES | June 16, 2026

AGAPI-Agents: An Open-Access Agentic AI Platform for Accelerated Materials Design on AtomGPT.org

[Jaehyung Lee](#), [Justin Ely](#), [Kent Zhang](#), [Akshaya Ajith](#), [Charles Rhys Campbell](#), and [Kamal Choudhary](#)*

[Home](#) > [Integrating Materials and Manufacturing Innovation](#) > Article

ChatGPT Material Explorer: Design and Implementation of a Custom GPT Assistant for Materials Science Applications

Brief Communication | Published: 09 July 2025

Volume 14, pages 276–283, (2025) [Cite this article](#)

Note: Most of the Agents use commercial products except AGAPI



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AtomGPT.org API (AGAPI)

Login with Email/Gmail/GitHub

Agentic AI Meets Materials Design

Query 100,000+ materials, predict properties with ML, build interfaces and grain boundaries, analyze spectra, and fold proteins, all from a chat UI or any LLM client.

Try it free →

Explore 50+ apps

Connect Claude / ChatGPT

Sign in

Reference: [1], [2]

Email

kchoudh2@jhu.edu

Password

.....

Sign in

Don't have an account? [Sign up](#)

or



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Specialized Features:

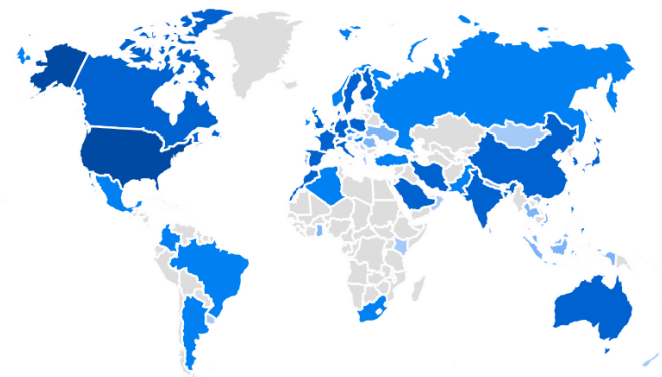
- 8+ Open Source LLMs, 50+ Apps
- Search materials database
- AI/ML property predictions
- Machine learning force-fields
- Protein structure analysis
- X-ray pattern to structure conversion
- Literature search
- Web search for real-time information,...
- Chat data NOT used for training



Technical Features:

- Function calling & tool use
- Model Context Protocol (MCP)

Worldwide usage
Currently ~10000 users



Learn more at GitHub, YouTube
<https://github.com/atomgptlab/agapi>

AtomGPTLab

Dr. Kamal Choudhary - 1 / 17



AI That Predicts Which
Materials Capture CO₂ From Air
Dr. Kamal Choudhary



XRD Suite: From Diffraction
Pattern to Crystal Structure in...
Dr. Kamal Choudhary



SuperconGPT: Forward and
Inverse Design for...

26



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Jaehyung Lee, Justin Ely, Kent Zhang, Akshaya Ajith, Charles Rhys Campbell, and Kamal Choudhary*

Cite

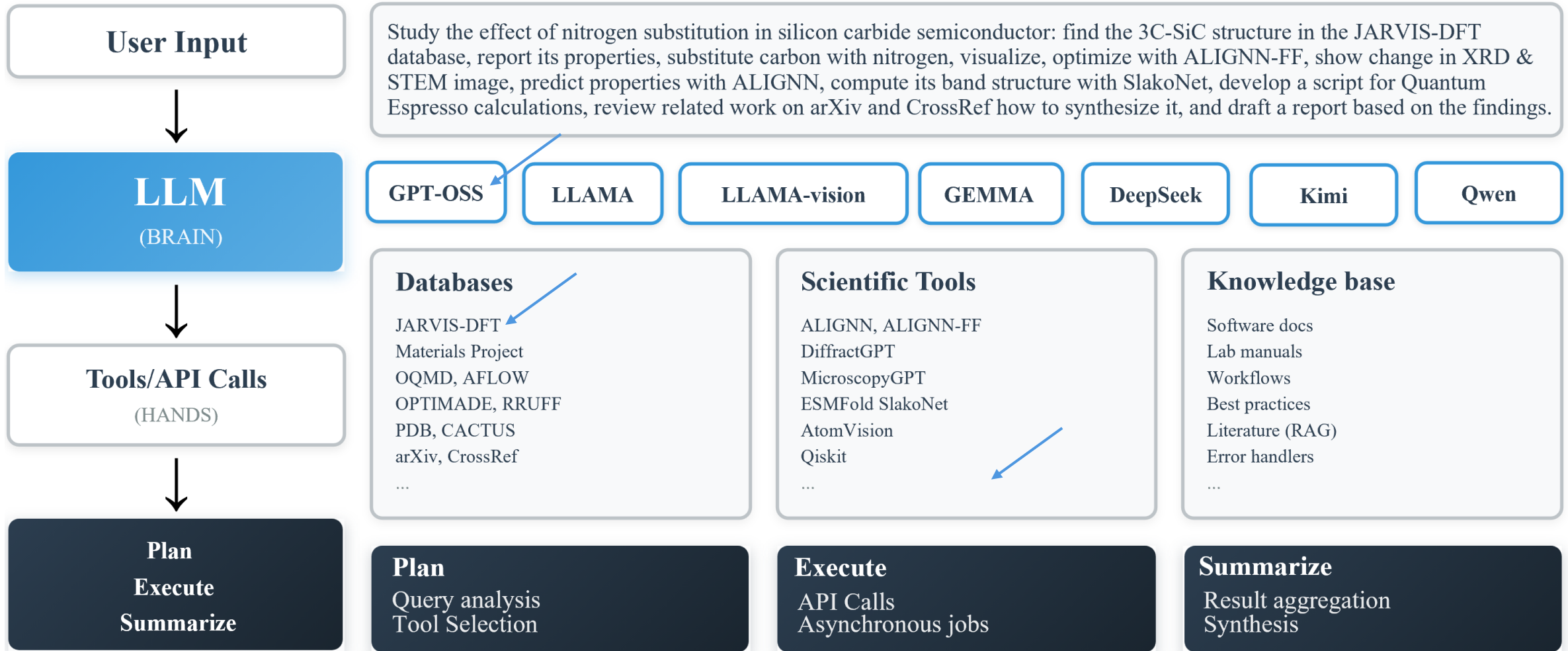
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Key Design Principles of AGAPI

AI Agents = Brain (LLM) + API (Hands) + Orchestration

<https://huggingface.com/> <https://atomgpt.org/apps> [OpenAI-Agents/MCP](#)



We discussed about LLMs in AtomGPT.org earlier, now let's discuss Tools/API Calls

Example 1: JARVIS-DFT+COD Databases

AtomGPT.org Materials Explorer

Advanced Search Across Materials Databases
Search through 100,000+ materials from JARVIS-DFT and COD databases with powerful filters:

- Element composition and chemical formulas
- Property ranges (band gap, T_c , spillage, exfoliation energy, and more)
- Crystal structure and space group
- Export results for further analysis

JARVIS-DFT
75,000+ DFT-calculated materials with comprehensive properties

COD
Crystallography Open Database — experimental structures

Basic Search

Chemical Formula:

Material ID:

Space Group:

Element Selection:

Contains ANY of

Element Selection Dropdown:

- MoS2
- LaB6
- AgS
- NaFe3Si2O6
- FeTe

Periodic Table:

H						He	
Li		B	C	N	O	F	Ne
Na	Mg	Al	Si	P	S	Cl	Ar

nature >npj computational materials > articles > article

Article | [Open access](#) | Published: 12 November 2020

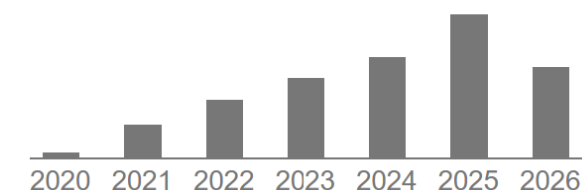
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https://atomgpt.org/materials_explorer



Example 2: OPTIMADE Explorer

AtomGPT.org

Apps About Docs GitHub

OPTIMADE Explorer

Query JARVIS-DFT via the OPTIMADE API: standardized access to various materials databases

OPTIMADE Filter

JARVIS-DFT 3D (~93K) elements HAS ANY "C","Si" **Query** **Reset**

EXAMPLES: elements HAS ANY C,Si,Ge elements HAS ALL C,Si formula = MoS2 anonymous = A28 nelements ≥ 3 nelements=2 AND nsites ≤ 4 Ti binaries

JARVIS Reference | OPTIMADE Standard | Andersen et al, Sci Data 8, 217 (2021) | atomgptlab/jarvis

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Article [Open access](#) Published: 12 August 2021

OPTIMADE, an API for exchanging materials data

[Casper W. Andersen](#), [Rickard Armiento](#), [Evgeny Blokhin](#), [Gareth J. Conduit](#), [Shyam Dwaraknath](#), [Matthew L. Evans](#), [Ádám Fekete](#), [Abhijith Gopakumar](#), [Saulius Gražulis](#), [Andrius Merkys](#), [Fawzi Mohamed](#), [Corey Oses](#), [Giovanni Pizzi](#), [Gian-Marco Rignanese](#), [Markus Scheidgen](#), [Leopold Talirz](#), [Cormac Toher](#), [Donald Winston](#), [Rossella Aversa](#), [Kamal Choudhary](#), [Pauline Colinet](#), [Stefano Curtarolo](#), [Davide Di Stefano](#), [Claudia Draxl](#), ... [Xiaoyu Yang](#) [+ Show authors](#)

[Scientific Data](#) **8**, Article number: 217 (2021) | [Cite this article](#)

[Save article](#)

15k Accesses | **114** Citations | **28** Altmetric | [Metrics](#)

https://atomgpt.org/optimade_explorer



Example 3: AI Property Prediction Model: ALIGNN

→ <https://atomgpt.org/alignn>

AtomGPT.org

Apps About Docs GitHub YouTube Spotify

ALIGNN Property Predictor

Predicted Properties (15 total):

- Energy & Stability
- Electronic & Band Gap
- Mechanical
- Dielectric & Piezo
- Topological
- Optical
- Magnetic
- Superconducting
- Interface

POSCAR Input:

```
System
1.0
3.2631502048902807 0.0 -0.0
0.0 3.2631502048902807 0.0
0.0 -0.0 3.2631502048902807
Ti Au
1 1
direct
0.5 0.5 0.5 Ti
0.0 0.0 0.0 Au
```

Predict Properties

Running ALIGNN predictions... (15 models)

Refs: PAPER [NPJ Comp Mat 7, 1 \(2021\)](#) | GITHUB [atomgptlab/alignn](#) | DOCS [AtomGPT Docs](#) | SCHOLAR [ALIGNN](#) — Google Scholar

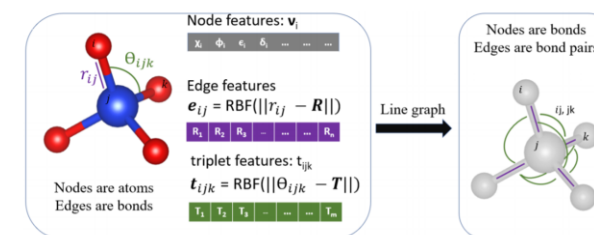
[nature](#) > [npj computational materials](#) > [articles](#) > [article](#)

Article | [Open access](#) | Published: 15 November 2021

Atomistic Line Graph Neural Network for improved materials property predictions

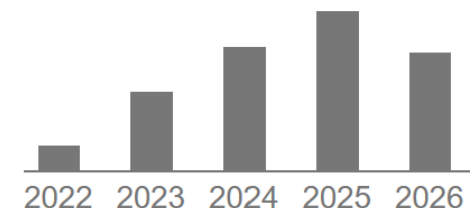
[Kamal Choudhary](#) & [Brian DeCost](#)

[npj Computational Materials](#) **7**, Article number: 185 (2021) | [Cite this article](#)



Atoms, Bonds are not enough
Need bond-angles

Cited by 755



<https://atomgpt.org/alignn>



Example 4: Foundational ALIGNN-FF Model

← → ↻ https://atomgpt.org/alignn_ff_dynamics ☆ 📄 📁 🔍

AtomGPT.org Apps About Docs GitHub YouTube Spotify

⚡ ALIGNN-FF Dynamics & Optimization

Real-time geometry optimization, molecular dynamics, and phonon bandstructure powered by ALIGNN force field

🔗 Geometry Optimization

🔥 Molecular Dynamics

🎵 Phonon Bandstructure

🌟 **Geometry Optimization:** Relax atomic positions and cell parameters to minimize energy. Uses ALIGNN-FF for fast, accurate force calculations. Max 100 atoms.

POSCAR Input:

```
System
1.0
5.0 0.0 0.0
0.0 5.0 0.0
0.0 0.0 5.0
Si
8
direct
0.25 0.75 0.25 Si
0.0 0.0 0.5 Si
0.25 0.25 0.75 Si
0.0 0.5 0.0 Si
0.75 0.75 0.75 Si
0.5 0.0 0.0 Si
0.75 0.25 0.25 Si
0.5 0.5 0.5 Si
```

Optimization Parameters:

OPTIMIZER
FIRE (Fast)

FORCE TOLERANCE (eV/Å)
0.05

MAX STEPS
100

☒ RELAX CELL

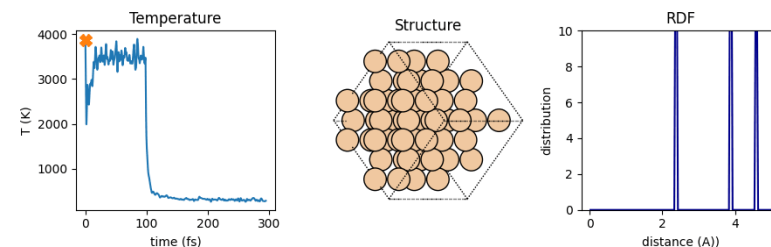
▶ Run Optimization

🗑 Clear

Creative Commons Attribution 3.0 Unported Licence
DOI: [10.1039/D2DD00096B](https://doi.org/10.1039/D2DD00096B) (Paper) *Digital Discovery*, 2023, 2, 346-355

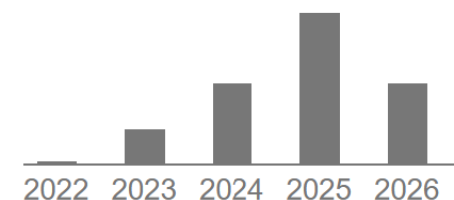
Unified graph neural network force-field for the periodic table: solid state applications

Kamal Choudhary ^{ab}, Brian DeCost ^c, Lily Major ^{de}, Keith Butler ^e, Jeyan Thiyagalingam ^e and Francesca Tavazza ^c



Not just static but dynamic properties

Cited by 148



https://atomgpt.org/alignn_ff_dynamics

Example 5: Foundational Slater-Koster Model: SlaKoNet

AtomGPT.org

Apps About Docs GitHub YouTube

SlaKoNet Band Structure Calculator

POSCAR Input:

```
MgB2
1.0
1.5367454354207384 -2.661721209337192 0.0
1.5367454354207384 2.661721209337192 0.0
0.0 0.0 3.5146401326518166
Mg B
1 2
Cartesian
0.0 0.0 0.0
1.53675 -0.8872417744799828 1.75732
1.53675 0.8872417744799828 1.75732
```

Band Structure + DOS 3D Band Surface Fermi Surface

Band Structure & DOS

Energy min (eV) Energy max (eV)

-8 8 Calculate

Refs: PAPER J Phys Chem Lett 16, 11109 (2025) — SlaKoNet | GITHUB atomgptlab/slakonet | DOCS AtomGPT Docs

The Journal of Physical Chemistry Letters > Vol 16/Issue 43 > Article

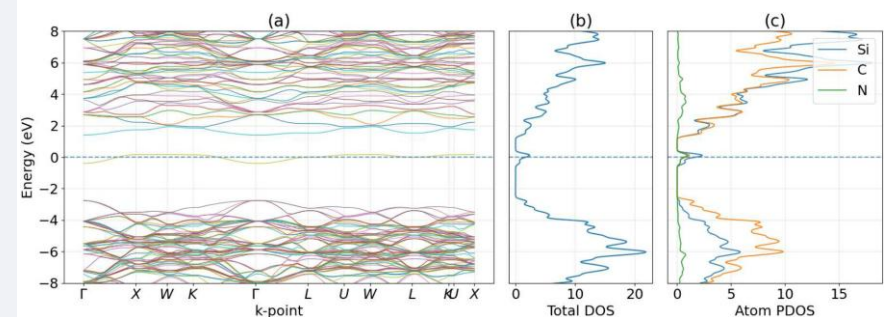
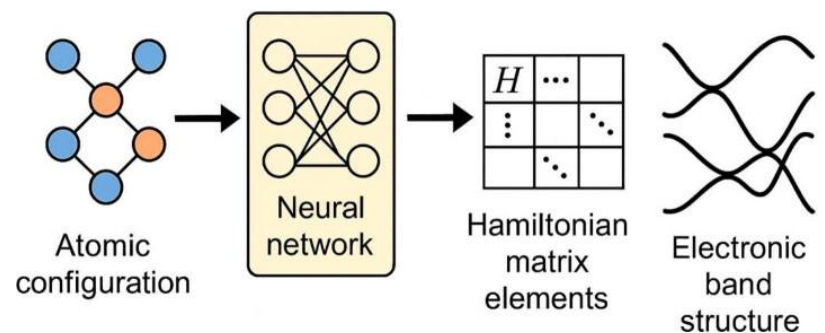
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PHYSICAL INSIGHTS INTO MATERIALS AND MOLECULAR PROPERTIES | October 17, 2025

SlaKoNet: A Unified Slater-Koster Tight-Binding Framework Using Neural Network Infrastructure for the Periodic Table

Kamal Choudhary*



Atoms, Bonds, Angles are not enough
Need electronic orbitals

<https://atomgpt.org/slakonet>



Example 6: Superconductor App

AtomGPT.org

Apps About Docs GitHub YouTube

SuperconGPT — Superconductor Tools

Crystal Generator Database Search Tc Predictor

SuperconDB Search: DFT-based BCS superconductor database for 3D (1058) and 2D (161) materials at ambient conditions. Click periodic table elements to build your query, then click Search.

SELECT ELEMENTS

Selected Elements

<https://atomgpt.org/supercon>

nature > npj computational materials > articles > article

Article | [Open access](#) | Published: 22 November 2022

Designing high- T_c superconductors with BCS-inspired screening, density functional theory, and deep-learning

[Kamal Choudhary](#) & [Kevin Garrity](#)

[npj Computational Materials](#) **8**, Article number: 244 (2022) | [Cite this article](#)

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The Journal of Physical Chemistry Letters > Vol 15/Issue 27 > Article

Cite Share Jump to

PHYSICAL INSIGHTS INTO MATERIALS AND MOLECULAR PROPERTIES | June 27, 2024

AtomGPT: Atomistic Generative Pretrained Transformer for Forward and Inverse Materials Design

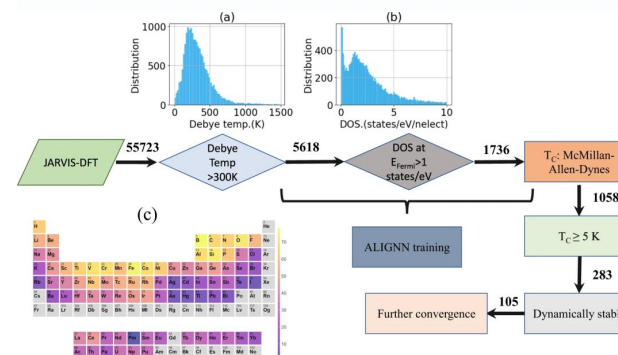
[Kamal Choudhary*](#)

Hugging Face

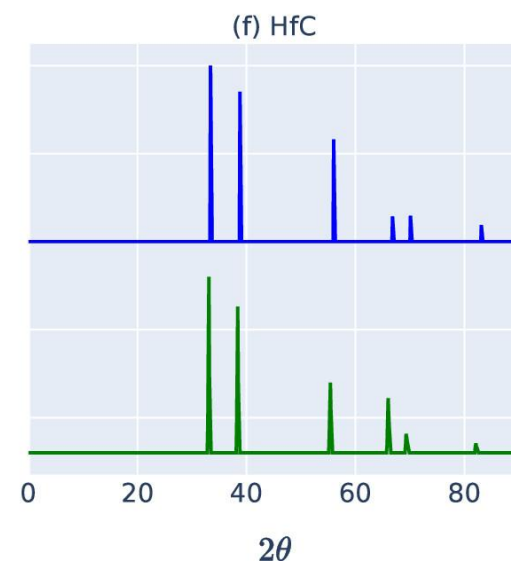
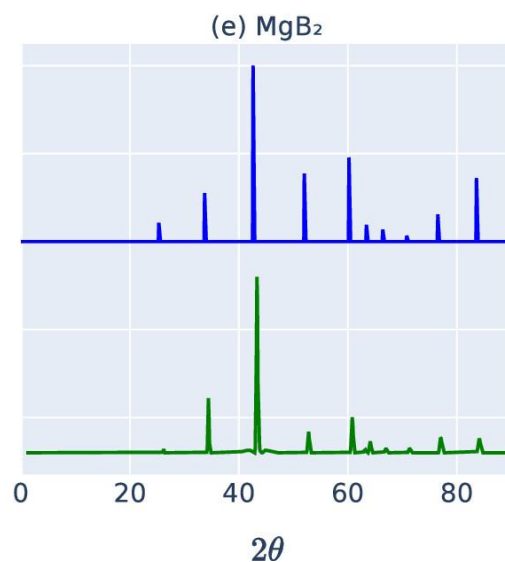
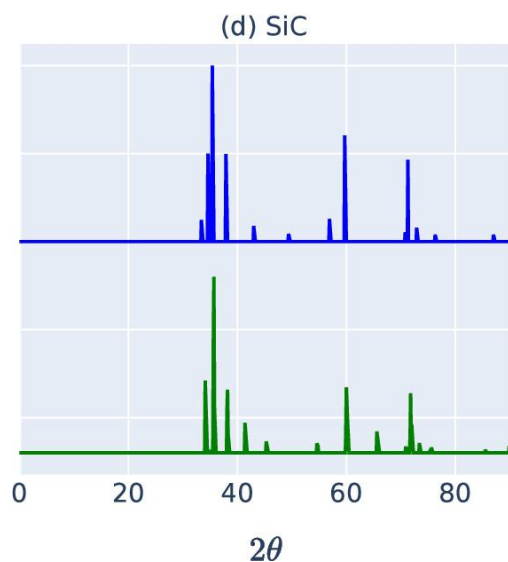
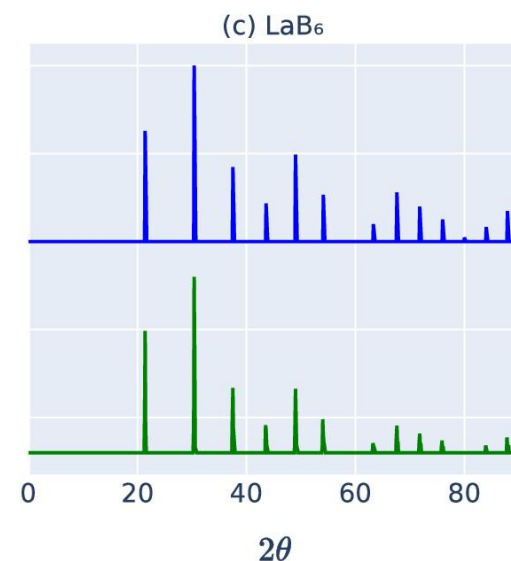
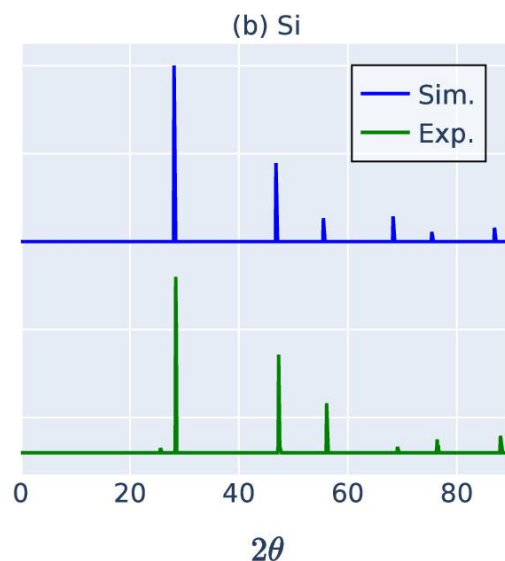
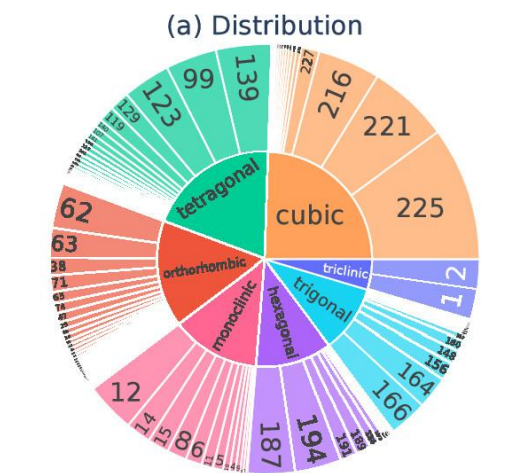
Search models, datasets, users...

[knc6/atomgpt_mistral_tc_supercon](#) like 6

Text Generation PEFT Safetensors English chemistry text-generation



PXRD Database for Training



```
class XRD(object):
    """Constryct an XRD class."""

    def __init__(
        self,
        wavelength=1.54184,
        thetas=[0, 180],
        two_theta_array=[],
        dhkl_array=[],
        intensity_array=[],
        scaling_factor=100,
        two_theta_tol=1e-5,
        intensity_tol=0.5,
        max_index=5,
    ):
        """Initialize class with incident wavelength."""
        self.wavelength = wavelength
        self.min2theta = np.radians(thetas[0])
        self.max2theta = np.radians(thetas[1])
        self.thetas = thetas
        self.two_theta_array = two_theta_array
        self.dhkl_array = dhkl_array
        self.intensity_array = intensity_array
        self.two_theta_tol = two_theta_tol
        self.intensity_tol = intensity_tol
```



Example 7: AI for XRD, DiffractGPT WebApp

AtomGPT.org

XRD Suite

X-ray diffraction pattern analysis, structure identification, and pattern generation

XRD Analyzer XRD Generator

XRD Analyzer: Identify crystal structures from experimental XRD patterns. Pattern matching against JARVIS-DFT + COD databases, AI-powered DiffractGPT, optional Rietveld refinement (GSAS-II or BGMN), preferred orientation correction, and ML property predictions.

[DiffractGPT](#) · [AGAPT-XRD](#)

INPUT PARAMETERS

CHEMICAL FORMULA / ELEMENTS

Formula (SiO2) or comma-separated elements (SiO2)

XRD PATTERN DATA

21.28	100
30.32	95
37.38	90
43.44	25
48.88	55
53.92	30
63.16	15
67.48	30
71.68	22
75.78	17

2θ and intensity, one pair per line. For Rietveld, use full continuous scan (1-200 pts).

WAVELENGTH (Å)

METHOD Pattern Matching

POST-PROCESSING

☐ Rietveld Refinement

RESULTS

Enter XRD data and click Run Analysis.

The Journal of Physical Chemistry Letters > Vol 16/Issue 8 > Article

Open Access

PHYSICAL INSIGHTS INTO MATERIALS AND MOLECULAR PROPERTIES | February 20, 2025

DiffractGPT: Atomic Structure Determination from X-ray Diffraction Patterns Using a Generative Pretrained Transformer

Kamal Choudhary*

[Open PDF](#) [Supporting Information \(1\)](#) [FIND IT](#)

Downloads last month
240,592

Inference Providers NEW

[Text Generation](#)

This model isn't deployed by any Inference Provider. [1](#) Ask for provider support

[Model tree for knc6/diffractgpt_mistral_chemical_formula](#)

Think DiffractGPT as
AlphaFold for MSE

Similar to mainstream
DeepSeek v3 downloads

<https://atomgpt.org/xrd>

AtomVision

MACHINE LEARNING AND DEEP LEARNING | March 1, 2023

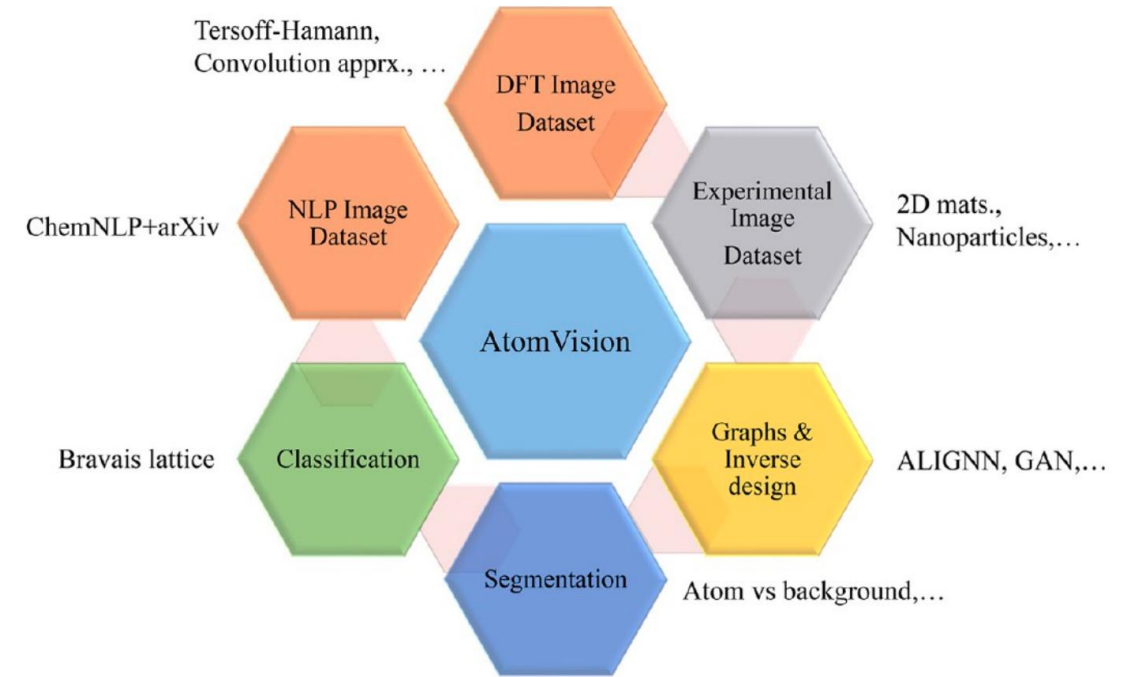
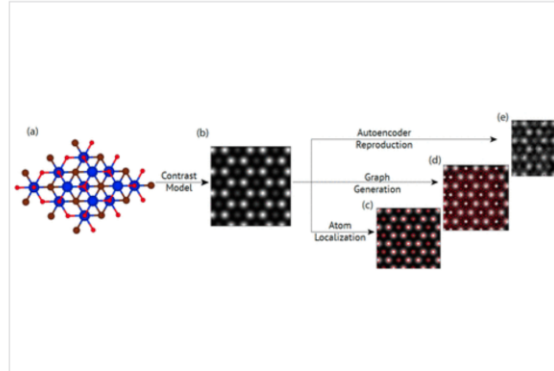
AtomVision: A Machine Vision Library for Atomistic Images

Kamal Choudhary*, Ramya Gurunathan, Brian DeCost, and Adam Biacchi

[Other Access Options](#)

Abstract

Computer vision techniques have immense potential for materials design applications. In this work, we introduce an integrated and general-purpose AtomVision library that can be used to generate and curate microscopy image (such as scanning tunneling microscopy and scanning transmission electron microscopy) data sets and apply a variety of machine learning techniques. To demonstrate the applicability of this library, we (1) establish an atomistic image data set of about 10 000 materials with large structural and chemical diversity, (2) develop and compare convolutional and atomistic line graph neural network models to classify the Bravais lattices, (3) demonstrate the application of fully convolutional neural networks using U-Net architecture to pixelwise classify atom versus background, (4) use a



Various tasks with AtomVision

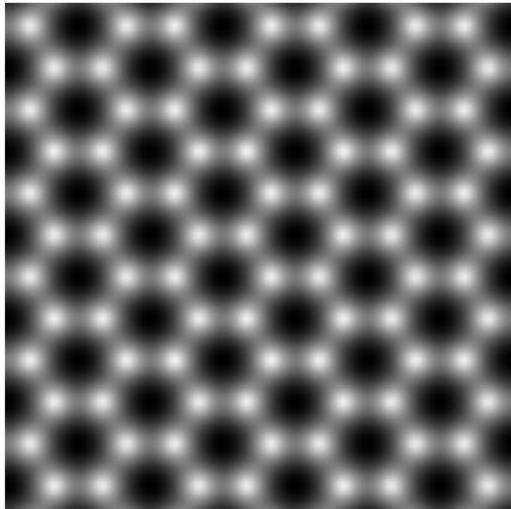
<https://github.com/atomgptlab/atomvision>



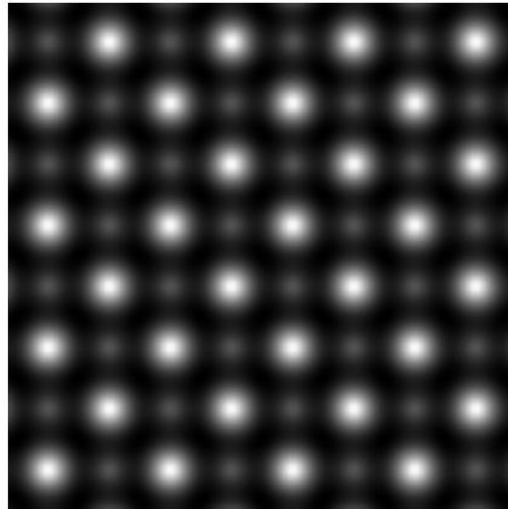
Scanning Transmission Electron Microscope Image

- Convolution approximation: accurate for **thin films** mainly (here 2D mats.)
- Based on Rutherford scattering model

$$I(\mathbf{r}) = R(\mathbf{r}, Z) \otimes \text{PSF}(\mathbf{r}), \quad R(\mathbf{r}, Z) = \sum_{i=1}^N Z_i^{1.7} \delta(\mathbf{r} - \mathbf{r}_i)$$



C: JVASP-667

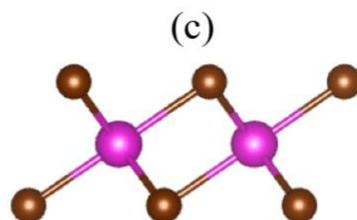
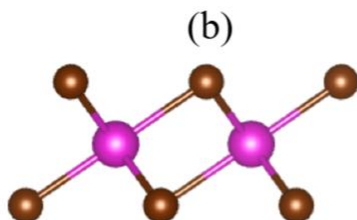
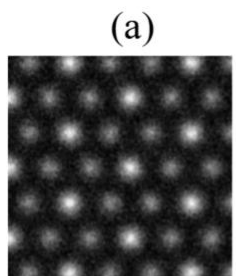


FeTe: JVASP-6667

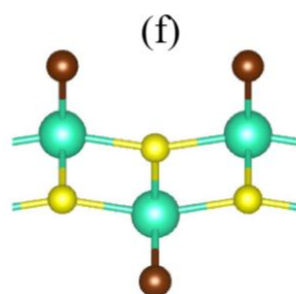
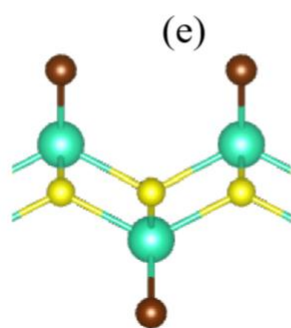
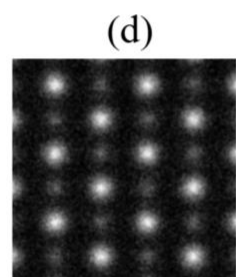


PPdSe: JVASP-6316

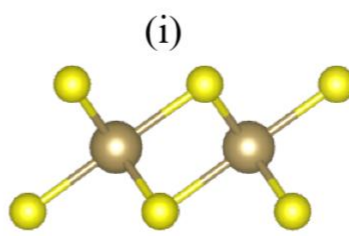
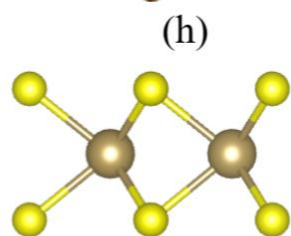
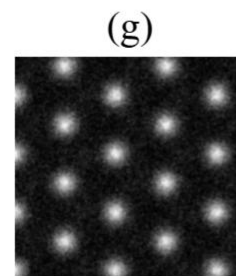
MicroscopyGPT: AtomGPT for Images



CdBr₂ (P $\bar{3}$ m1),
JVASP-76512,
RMSE (As Gen.): 0.00
RMSE (ALIGNN-FF):
0.00



LuSBr (Pmmn),
JVASP-6205,
RMSE (As Gen.): 0.10,
RMSE (ALIGNN-FF):
0.05



TaS₂ (P $\bar{6}$ m2),
JVASP-6070,
RMSE (As Gen.): 0.17
RMSE (ALIGNN-FF):
0.15

Prop.	Min.	Max.	KLD	EMD
a (Å)	2.38	14.398	0.06	1.67
b (Å)	2.51	15.64	0.07	0.91
γ (°)	60.0	120.0	0.02	0.54
Spg.	1	191	0.72	3.24
Weight (AMU)	24.82	2958.62	0.11	1.08

THE JOURNAL OF
PHYSICAL CHEMISTRY
LETTERS
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Open Access

Letter

MicroscopyGPT: Generating Atomic-Structure Captions from Microscopy Images of 2D Materials with Vision-Language Transformers

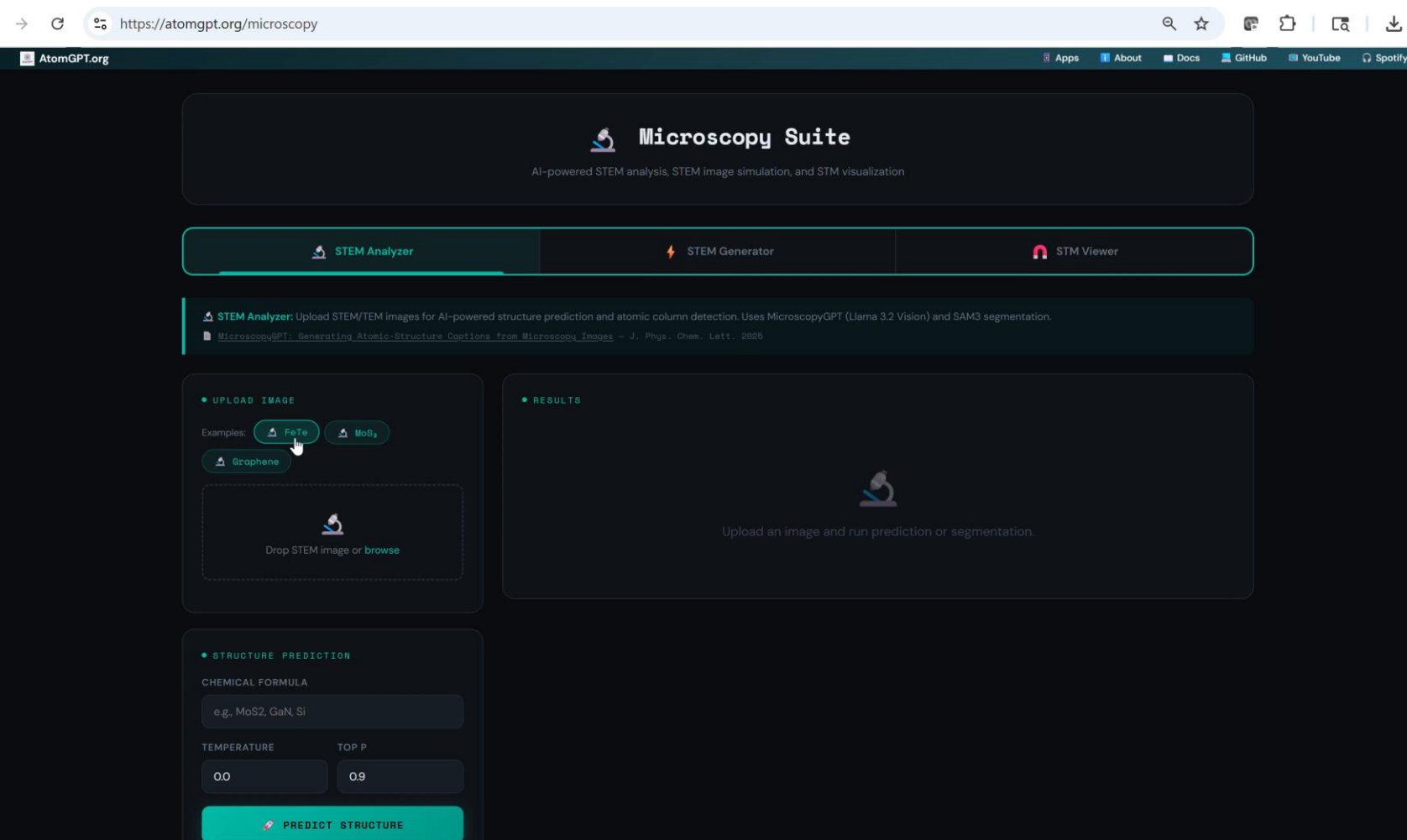
Kamal Choudhary*

Cite This: *J. Phys. Chem. Lett.* 2025, 16, 7028–7035

Read Online



Example 8: MicroscopyGPT, Image Segmentation



The Journal of Physical Chemistry Letters > Vol 16/Issue 27 > Article

Open Access

PHYSICAL INSIGHTS INTO MATERIALS AND MOLECULAR PROPERTIES | July 1, 2025

MicroscopyGPT: Generating Atomic-Structure Captions from Microscopy Images of 2D Materials with Vision-Language Transformers

Kamal Choudhary*

Hugging Face

Search models, datasets, users...

knc6/microscopy_gpt_llama3.2_vision_11b like 0

Text Generation PEFT Safetensors Transformers PyTorch English chemistry

License: mit

Model card Files and versions xet Community Settings

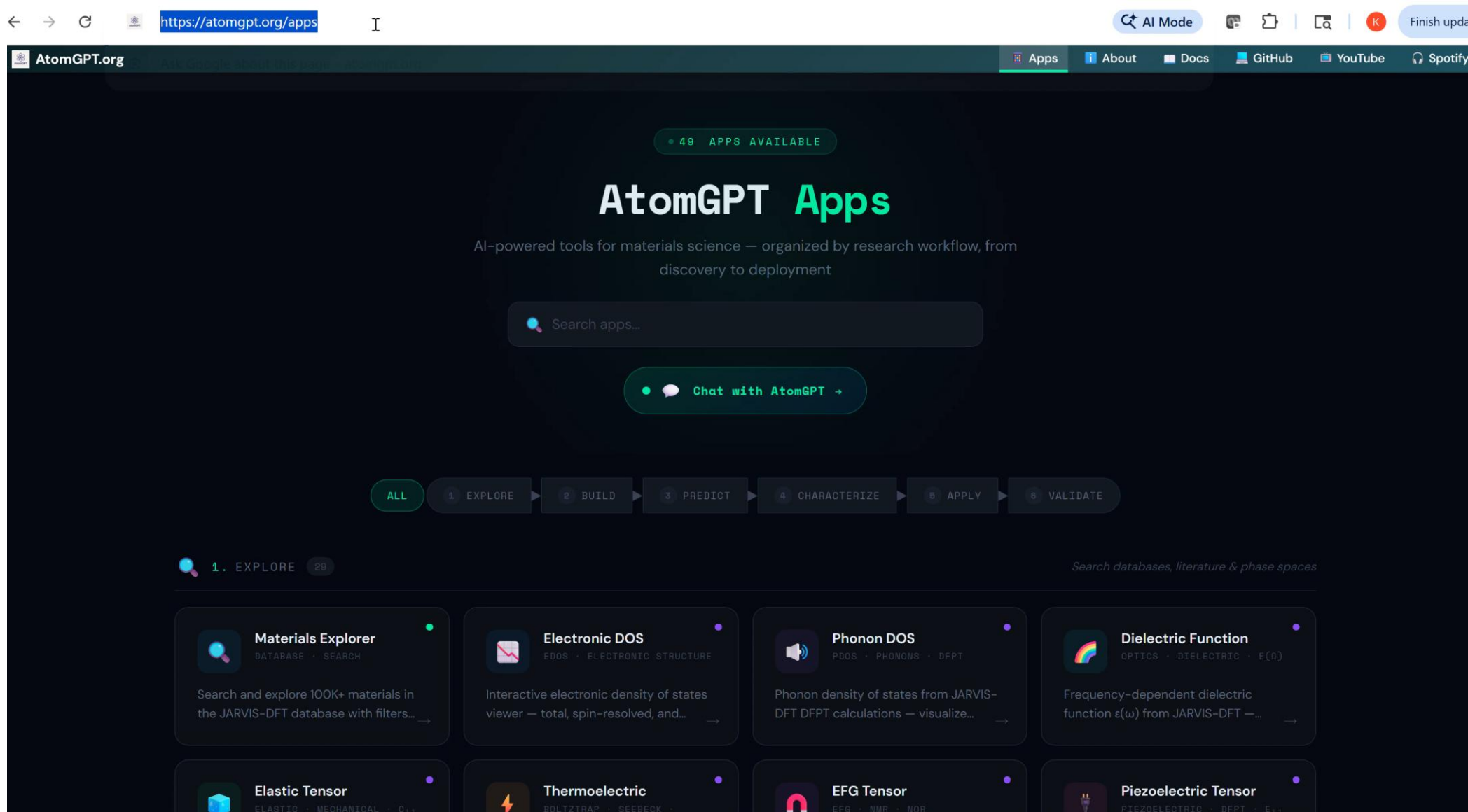
MicroscopyGPT: Generating Atomic-Structure Captions from Microscopy Images of 2D Materials with Vision-Language Transformers

Image Segmentation with Meta-SAM3

<https://atomgpt.org/microscopy>



50+ Domain Specific APIs as WebApps



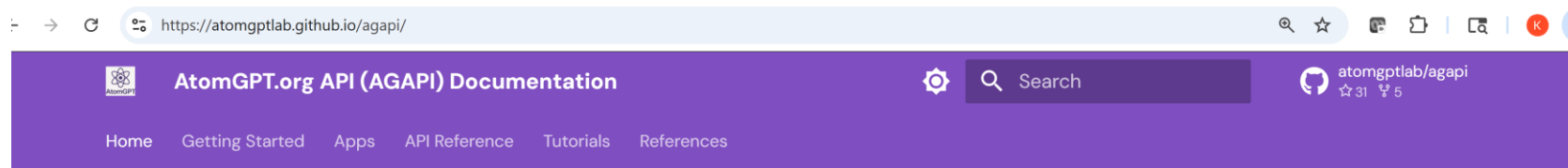
The screenshot shows the AtomGPT.org website interface. At the top, the browser address bar displays <https://atomgpt.org/apps>. The website header includes the AtomGPT.org logo and navigation links for Apps, About, Docs, GitHub, YouTube, and Spotify. A green badge indicates "49 APPS AVAILABLE". The main heading is "AtomGPT Apps", followed by the tagline "AI-powered tools for materials science — organized by research workflow, from discovery to deployment". A search bar labeled "Search apps..." and a "Chat with AtomGPT" button are present. A workflow navigation bar shows steps: ALL, 1. EXPLORE, 2. BUILD, 3. PREDICT, 4. CHARACTERIZE, 5. APPLY, and 6. VALIDATE. The "1. EXPLORE" section is active, with a sub-header "Search databases, literature & phase spaces". Below this, eight application cards are displayed in a grid:

- Materials Explorer**: DATABASE · SEARCH. Search and explore 100K+ materials in the JARVIS-DFT database with filters...
- Electronic DOS**: EDOS · ELECTRONIC STRUCTURE. Interactive electronic density of states viewer — total, spin-resolved, and...
- Phonon DOS**: PDOS · PHONONS · DFT. Phonon density of states from JARVIS-DFT DFT calculations — visualize...
- Dielectric Function**: OPTICS · DIELECTRIC · $\epsilon(\omega)$. Frequency-dependent dielectric function $\epsilon(\omega)$ from JARVIS-DFT —...
- Elastic Tensor**: ELASTIC · MECHANICAL · Q. ...
- Thermoelectric**: BOLTZTRAP · SEEBECK · ...
- EFG Tensor**: EFG · NMR · NQR · ...
- Piezoelectric Tensor**: PIEZOELECTRIC · DFT · e_{ij} · ...

<https://atomgpt.org/apps>



API Tools Documentation



Home

AtomGPT.org API(AGAPI) Documentation

Welcome to the documentation for [AtomGPT.org](https://atomgptlab.github.io/agapi/) API – a comprehensive platform with **50+ web applications** for materials science, powered by the JARVIS ecosystem.

Explore

Browse 76K+ materials, proteins, polymers, surfaces, and more from JARVIS-DFT and other databases.

→ [Explore Apps](#)

Build

Visualize structures, build heterostructures, design workflows, and write AI-assisted code.

→ [Build Apps](#)

Predict

ALIGNN, SlakoNet, ALIGNN-FF, AtomGPT — predict properties with GNNs, MLFF, TB, and quantum algorithms.

→ [Predict Apps](#)

Characterize

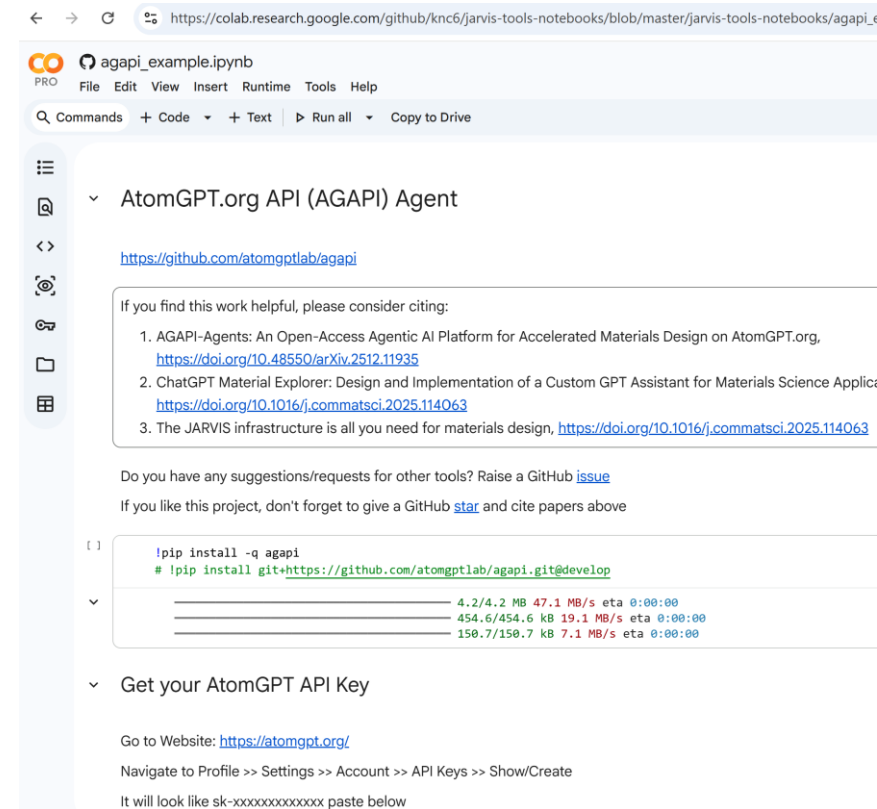
Match XRD patterns with DiffractGPT, analyze Raman spectra, identify structures from microscopy images.

→ [Characterize Apps](#)

Table of contents

Quick Links

Citation



<https://atomgptlab.github.io/agapi/>

https://colab.research.google.com/github/knc6/jarvis-tools-notebooks/blob/master/jarvis-tools-notebooks/agapi_example.ipynb



Example Query for Database Search

```
ag5=agent.query_sync("How many materials have Tc_supercon data?",render_html=True)
```

Answer:

There are **1,058** materials in the JARVIS-DFT database that have a recorded superconducting transition temperature ($T_{c_supercon}$).

[nature](#) > [npj_computational materials](#) > [articles](#) > [article](#)

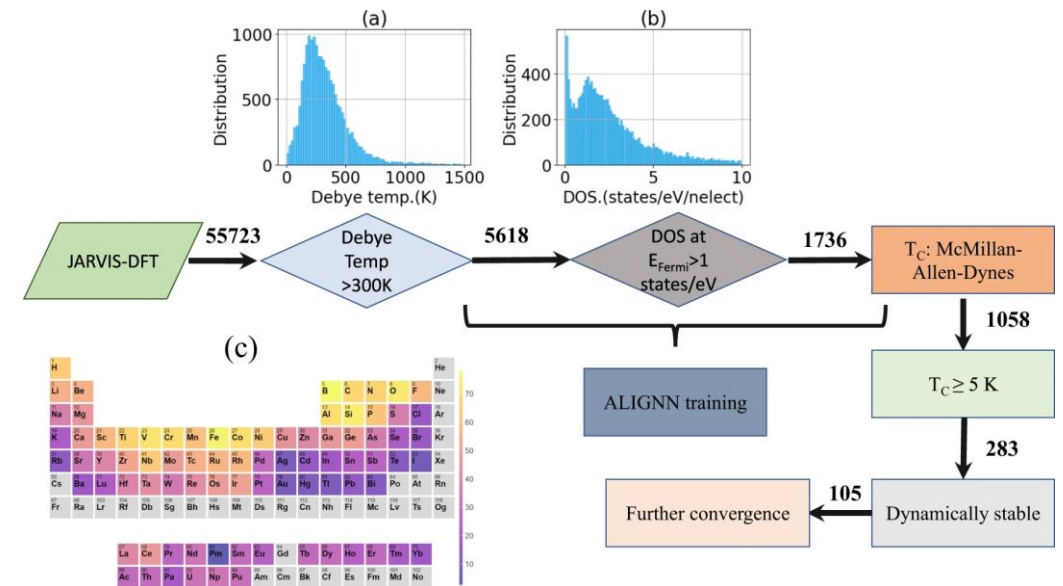
Article | [Open access](#) | Published: 22 November 2022

Designing high- T_C superconductors with BCS-inspired screening, density functional theory, and deep-learning

[Kamal Choudhary](#) & [Kevin Garrity](#)

[npj Computational Materials](#) **8**, Article number: 244 (2022) | [Cite this article](#)

12k Accesses | 69 Citations | 9 Altmetric | [Metrics](#)



If else statements vs. agentic search

Example Semiconductor Design: Multi-Agents/Prompts

```
from agapi.agents import AGAPIAgent
```

```
ag25 = agent.query_sync("""
```

1. Find all GaN materials in the JARVIS-DFT database
2. Get the POSCAR for the most stable one (lowest formation energy)
3. Make a 2x1x1 supercell
4. Substitute one Ga with Al
5. Generate its powder XRD pattern using Cu K-alpha radiation
6. Report the top 10 strongest peaks
7. Optimize structure with ALIGNN-FF
8. Predict properties with ALIGNN
9. Calculate band structure with SlakoNet
10. Then create a summary table showing:
 - Formation energy (ALIGNN prediction)
 - Bandgap MBJ (ALIGNN prediction)
 - Bandgap OptB88vdW (ALIGNN prediction)
 - Bandgap (SlakoNet calculation)
 - Bulk modulus (ALIGNN prediction)
 - Shear modulus (ALIGNN prediction)
 - XRD peaks

```
Display all values clearly.
```

```
""", auto_display_images=True, render_html=True, verbose=True, max_cont
```

The Journal of Physical Chemistry Letters > Vol 16/Issue 43 > Article

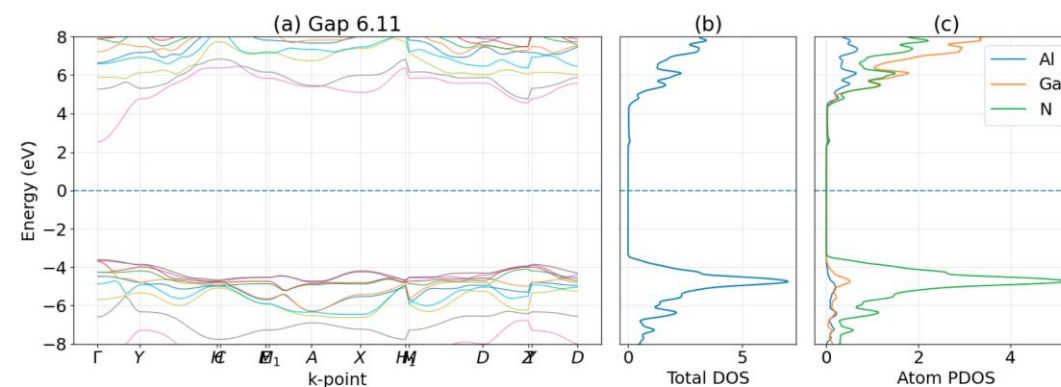
Subscribed

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PHYSICAL INSIGHTS INTO MATERIALS AND MOLECULAR PROPERTIES | October 17, 2025

SlakoNet: A Unified Slater-Koster Tight-Binding Framework Using Neural Network Infrastructure for the Periodic Table

Kamal Choudhary*



✓ Displayed band structure image

Al-doped GaN (AlGa₃N₄) - 2 × 1 × 1 supercell (1 Al, 3 Ga, 4 N)



Example XRD Pattern

```
from agapi.agents import AGAPIAgent
```

```
ag27 = agent.query_sync("""
```

```
1. Find GaN in JARVIS-DFT
```

```
2. Get the most stable structure
```

```
3. Generate its powder XRD pattern using Cu K-alpha radiation
```

```
4. Report the top 10 strongest peaks
```

```
""", verbose=True, render_html=True)
```

```
[Iteration 1/15]
```

```
  Calling: query_by_elements
```

```
[Iteration 2/15]
```

```
  Calling: query_by_jid<|channel|>
```

```
[Iteration 3/15]
```

```
  Calling: generate_xrd_pattern
```

```
  [Truncated result from 36676 to 896 chars]
```

```
[Iteration 4/15]
```

```
**GaN - Most Stable Structure (JARVIS-DFT)**
```

```
- **JARVIS ID:** **JVASP-30**
```

```
- **Crystal System:** Hexagonal (space-group  $P6_3mc$ )
```

```
- **Lattice Parameters:**
```

```
  -  $a^*$  = 3.211 Å (derived from the POSCAR)
```

```
  -  $c^*$  = 5.240 Å
```

```
- **Formation Energy:** -0.571 eV / atom (thermodynamically the lowest-energy polymorph i
```

```
- **Bandgap (MBJ):** 3.08 eV
```

```
**Powder XRD Pattern (Cu K $\alpha$ ,  $\lambda$  = 1.54184 Å)**
```

```
The `generate_xrd_pattern` tool returned 8 distinct peaks within the 0-90° 2 $\theta$  range.
```

```
Below are the **top 10 strongest peaks** (sorted by relative intensity). Since only 8 pea
```

Rank	2 θ (°)	Intensity (relative)	d-spacing (Å)
1	36.6	1.00	2.455
2	32.2	0.44	2.780
3	34.2	0.36	2.622
4	47.7	0.21	1.907
5	57.4	0.30	1.605
6	62.8	0.31	1.480
7	68.5	0.25	1.370
8	70.0	0.15	1.344



Outline

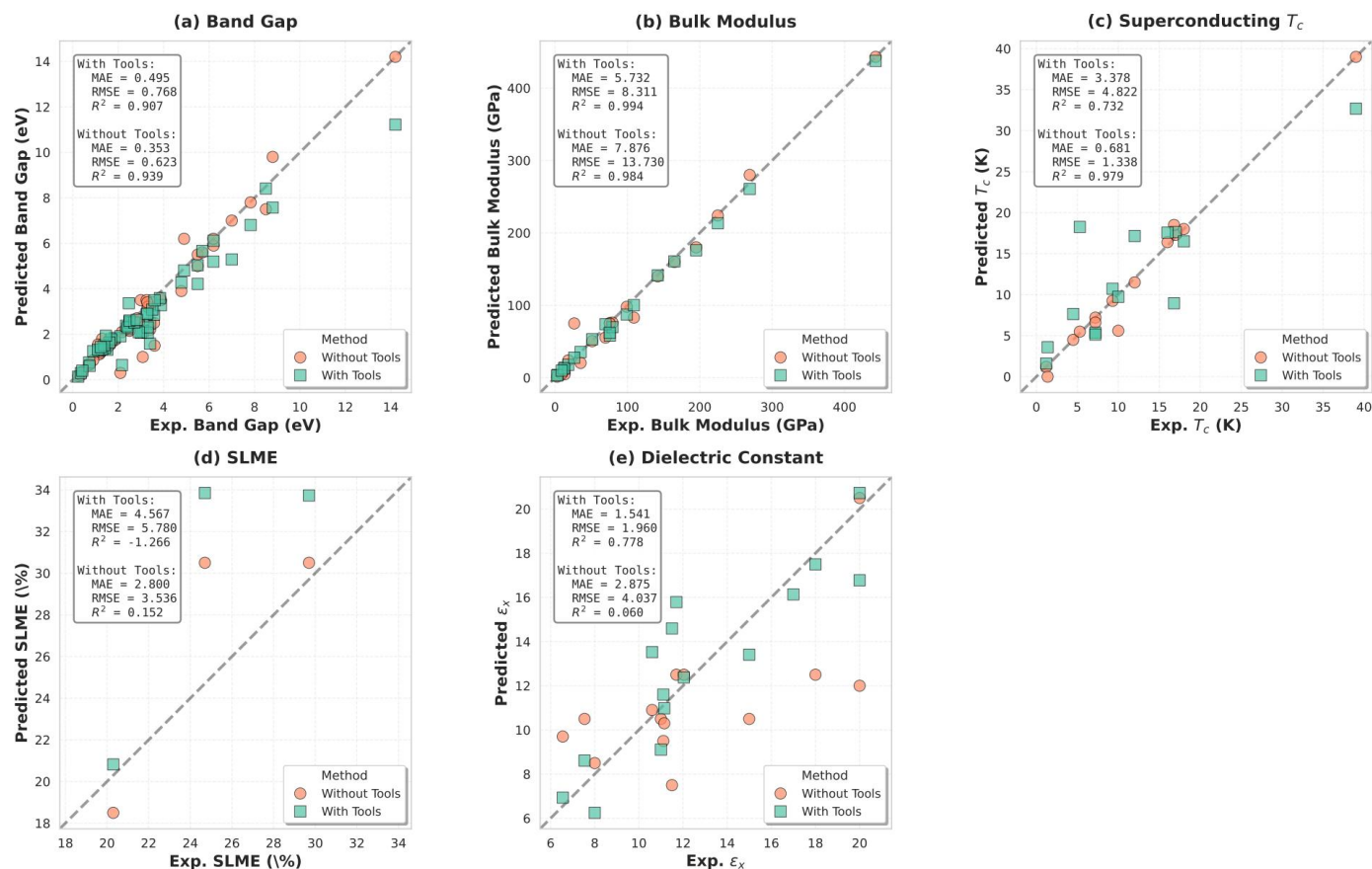
Overview of Agentic AI

ChatGPT Material Explorer

AtomGPT.org API Demos

Benchmarking & Hackathon

Agentic AI Benchmarking on Well-known Materials



Findings suggest that tool augmentation is not universally beneficial, its effectiveness depends critically on three factors:

- (1) database quality and coverage for the target property,
- (2) standardization of computational protocols used to generate database entries, and
- (3) whether the material and its properties are well-documented in existing literature.

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Jaehyung Lee, Justin Ely, Kent Zhang, Akshaya Ajith, Charles Rhys Campbell, and Kamal Choudhary*

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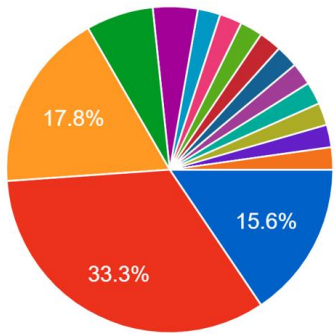


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Agentic AI for Science (AAI4Science) Hackathon 2025

Date: Thursday, November 6, 2025
Location: Bloomberg Student Center, Room 204 (Hybrid)
Registration: [Register here](#)
Colab Notebook: [Open in Colab](#)
Submission Form: [Google Form](#)
AtomGPT.org API (AGAPI): [GitHub Repository](#)

Stay tuned for 2026 AAI4Science hackathon at JHU+virtual!
Signup at atomgpt.org



Let's collaborate!

Laboratories

NIST:

F. Tavazza, B. DeCost, K. Garrity, C. Campbell, D. Wines, A. Biacchi, A. Hight Walker, S. Krylyuk, C. Camp Jr.

ORNL:

R. Vasudevan, D. Zhang, B. Sumpter, V. Sharma

ANL:

N. Anand

PNNL:

M. Ziatdinov

SLAC:

A. Mehta

LLNL:

T. Frolov

Industry & Other

GE Aerospace:

G. Pilania

FDA:

R. Beams

Universities

Northwestern:

A. Agrawal, A. Choudhary, W. Liao, D. Jha

UMD:

A. Kusne, I. Takeuchi, V. Stanev

UTK/UT:

S. Kalinin, T. Liang, L. Vlcek

UF:

S. Phillpot, R. Hennig

JHU:

Y. Cheng

Penn State:

S. Sinnott

Toronto:

J. Hattrick-Simpers

Rutgers:

K. Rabe, D. Vanderbilt

GMU:

P. Vora, K. Sauer, I. Mazin

Virginia Tech:

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Universities (cont'd)

OSU:

A. Asthagiri, X. Nie

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H. Zhuang

Texas A&M:

X. Qian

Missouri S&T:

A. Chernatynskiy

IIT Hyderabad:

A. Goyal

Central South:

J. Jiang, P. Wang

UNICAMP:

A. Fontes da Fonseca

U. Antwerpen:

M. Bercx, D. Lamoen

NCSU:

E. Bucholz

PSI:

M. Bercx



Our team



Jayhuyng Lee C. Rhys Campbell



Justin Ely



Kent Zhang



Akshya Ajith



Shrijani
Buruganahalli



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(USNA)

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drkamal@jhu.edu

